NIKHIL DEEKONDA

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EDUCATION

Yeshiva University, Katz School of Health and Science, NY, USA Sep 2023 - May 2025
Masters of Science in Artificial Intelligence **CGPA** 3.9

WORK EXPERIENCE

Lumamind LLC, North Hollywood, CA May 2025 - Present
Agentic AI Engineer

- Designed and implemented multi-agent orchestration frameworks using AWS Bedrock for real-time collaboration among persona-based agents in patient engagement. This includes creating agent-driven workflows that integrate retrieval-augmented generation (RAG) for personalized care recommendations and interventions.
- Led the fine-tuning of the Dia 1.6B TTS model on AWS SageMaker, optimizing it for low-latency voice interaction in a healthcare context.
- Fine-tuned DeepSeek-R1 for personalized clinician-patient conversations, enabling enhanced context-driven responses and reducing interaction time by refining conversation-based prompts. Currently setting up LangGraph to investigate whether agent latency can be reduced, ensuring seamless communication between dynamic agents.
- Developed dynamic prompting techniques using system attributes, enabling agents to fetch personalized patient data from secure backend services for real-time responses in medical scenarios.
- Applied evidence-based research techniques in RAG by integrating research papers filtered through MEDLINE and parsed using the Llama Parser to improve the accuracy and reliability of medical recommendations provided by the AI system.
- Architected real-time alerting systems using Server-Sent Events (SSE) to notify clinicians instantly of critical patient interactions, such as missed medications or relapse indicators, enhancing proactive decision-making.
- Designed and developed an iOS application (using Swift UI) enabling secure and conversational patient engagement through both text and voice modalities, integrated with backend AI systems
- Fine-tuned DeepSeek-R1 to adhere to clinical guidelines, ensuring that the AI system provides responses that are not only contextually relevant but also aligned with standard clinical practices.
- Implemented SOAP (Subjective, Objective, Assessment, Plan) formatting for consistent, structured, and easy-to-understand outputs for clinician-patient interactions, improving the communication of patient information.
- Developed and maintained a secure FastAPI backend hosted on AWS EC2 that integrates AWS Polly and Transcribe for real-time voice interactions and automatic transcription of patient interactions. These components are essential for ensuring smooth clinician-patient communication.
- Integrated AWS Cognito for secure authentication and HIPAA compliance across all data interactions, ensuring safe and private management of patient profiles and healthcare workflows.
- Currently developing LangChain-based agent workflows with LangGraph for multi-turn interactions, using dynamic prompts to optimize agent responses based on personalized patient histories for context-aware outputs.

LTIMindtree, Pune, Maharashtra, India Jul 2022 - Jul 2023
Senior Software Engineer

- Developed guardrails, data pipelines, and validation flows for ML/DL models including LLMs; fine-tuned models like GPT, Falcon, LLaMA-2, and Mistral for real-world use cases (PII removal, tone classification, toxic word detection) using LoRA and QLoRA techniques.
- Built a scalable platform to fine-tune LLMs with multiple techniques and implemented output validation for LLM agents utilizing frameworks such as AutoGen and LangGraph.

- Designed and executed complex Retrieval-Augmented Generation (RAG) workflows integrating LLMs with vector databases; developed hallucination detection systems leveraging Small Language Models (SLMs) and BERT-based models.
- Trained a proprietary 350M parameter language model from scratch and deployed fine-tuning, RAG, and validation workflows on AWS using services like EC2, S3, SageMaker, and Bedrock

National Highway Authority of India, Tirupati, Andhra Pradesh, India Jun 2021 - Aug 2021
Machine Learning Intern

- Applied various machine learning algorithms to NHAI road dataset of half a million instances, evaluating models based on R-squared values, successfully predicting road rutting factors, based on the best performing model.

PROJECTS

TaskFin: RAG-Based Financial Task Automation

- Designed and implemented a RAG-based system for end-to-end bill payment automation, leveraging LangChain agents and the LangGraph framework to orchestrate multi-turn interactions and ensure context-aware, accurate financial task management from natural language input.
- Developed a robust Orchestrator Agent to coordinate specialized sub-agents for Authentication, Financial Transactions, and State Management, using LangChain memory (buffer, entity, summary) and MCP servers to maintain persistent contextual state across interactions.
- Designed detailed PRDs (Product Requirement Documents) before initiating development, ensuring clear objectives and measurable outcomes for the project.
- Deployed the complete solution as a fully interactive Streamlit conversational app, enabling real-world RAG-driven task automation, secure data handling, and rapid prototyping of AI-powered fintech assistants.

LungAware: AI Lung Cancer Detection App

- Engineered a deep convolutional neural network (CNN) for early lung cancer detection, achieving 99.4% classification accuracy through advanced image preprocessing, transfer learning, and hyperparameter optimization.
- Developed cross-platform (Android/iOS) mobile applications with Swift (iOS) and Kotlin (Android), integrating TFLite-quantized models and Grad-CAM for explainable AI and transparent diagnostic support.
- Implemented a seamless user interface and cloud-based backend (AWS EC2) for automated inference, patient management, and result notification, ensuring compliance with healthcare privacy and security best practices.

Enhancing LLM Reliability: Automated Fact-Checking with Cross-Encoder Model

- Built and fine-tuned a Microsoft DeBERTa v3 large language model as a fact-checking cross-encoder, achieving an F1 score of 85% on the Facebook Fact Checking Dataset and significantly reducing hallucination in downstream NLP systems.
- Automated real-time fact-checking in conversational and document QA scenarios, increasing the credibility and accuracy of LLM-generated content for end-users.

Optimizing Visual Bird Sound Denoising with Deep Learning: A Segmentation Approach

- Designed a custom encoder-decoder deep neural network architecture for visual bird sound denoising, outperforming state-of-the-art models with an IoU of 66.24 on challenging real-world datasets.
- Integrated advanced image segmentation and noise reduction algorithms, demonstrating measurable improvements in ecological acoustic monitoring and bioacoustics research.

CERTIFICATIONS

- AWS Certified Machine Learning Engineer – Associate
- AWS Certified AI Practitioner